

Grade 1 Ontario Curriculum Algebra

Real-Life Math

Unit Overview: Real-Life Math

Grade: Grade 1 | **Strand:** Algebra

Target Expectations: C4

Learning Goal: Children will apply the mathematical modelling cycle (situation → question → data → model → predict → reflect) to a real-life situation.

Lesson	Lesson Title	Learning Goal
Day 1	What is the Sugar Bush?	I can describe a real-life situation and ask math questions about it.
Day 2	Counting Trees and Taps	I can collect data to answer a math question about a real-life situation.
Day 3	Predicting Sap Buckets	I can use data to predict an outcome.
Day 4	Reflecting and Refining	I can check my prediction against what really happens and adjust the model.
Day 5	My Real-Life Math	I can model a real-life situation from my own life.

Lesson 1: What is the Sugar Bush?

Goal: Modelling starts with a real-life situation. We notice it and ask MATH questions about it.

Expectation: C4

Learning Goal: "I can describe a real-life situation and ask math questions about it."

Materials: M1_paper_trees, M3_modelling_chart, M7_situation_cards, M10_sugar_bush_poster

1. Minds On (12 mins): Maple Welcomes the Class

What is a sugar bush? What happens there? Maple opens with the unit refrain: 'HOW MANY? HOW MUCH? HOW LONG? HOW TALL? — that's a math question, every one!' (Class echoes.)

Bring children to the carpet. Reveal Maple the Maple Tree mascot. 'Friends — meet Maple. Maple is a maple tree. Maple lives at the SUGAR BUSH. Today we'll think like math scientists about what happens there.'

2. Action (25 mins): Situation Stations

- Whole group: brainstorm 3 more sugar-bush questions.
- Send pairs to three stations (~7 min each).
- Stretch challenge at Station C: ask a question that requires PREDICTION (not just counting).
- Discuss: what makes a question 'mathematical'?
- Circulate to record formative evidence on the situation tracker.
- STRATEGY COMPARE: 2-3 pairs share their wonder-question lists.

3. Consolidation (13 mins): Maple's Question Brag

Discuss:

- What is a sugar bush?
- What's a math question?
- Why do we ask math questions about real life?
- What's one math question YOU asked today?
- Maple's preview: tomorrow Sap Sam shows us how to COUNT what we asked today (Lesson 2 — C4.1).

Exit: Maple's refrain — class chants 3x: "HOW MANY?"

Lesson 2: Counting Trees and Taps

Goal: We collect data — counts and measurements — so our model can answer the question.

Expectation: C4

Learning Goal: "I can collect data to answer a math question about a real-life situation."

Materials: M1_paper_trees, M4_data_table, M8_two_colour_counters, M3_modelling_chart

1. Minds On (12 mins): Sap Sam Counts the Trees

Sap Sam needs to know how many trees are in the sugar bush. How does he find out? Sap Sam opens: 'COUNT IT, WRITE IT, USE IT!' (Class echoes.)

Bring children to the carpet. Reveal Sap Sam. 'Friends — yesterday Maple welcomed us and we asked questions. Today Sap Sam needs ANSWERS. He needs to know how many trees are in the sugar bush.'

2. Action (25 mins): Data Stations

- Whole group: count more sets together.
- Send pairs to three stations (~7 min each).
- Stretch challenge at Station C: count and ALSO predict — 'If the class doubles, how many chairs?'
- Discuss the value of data tables.
- Circulate to record formative evidence on the data tracker.
- INVENT-AND-COMPARE: Pairs first record their Station A count however they choose (drawing, tally, number).

3. Consolidation (13 mins): Sap Sam's Data Brag

Discuss:

- What is data?
- How does data help us answer questions?
- Why use a table?
- What data did you collect today?

Exit: Sap Sam's refrain — class chants together 3x: "Sap Sam says: COUNT IT, WRITE IT, USE IT!" Each child reads their tally row aloud and uses the count in one sentence ('today I counted ____ ____, so ____').

Lesson 3: Predicting Sap Buckets

Goal: We use the data to make PREDICTIONS. If 1 tap fills 1 bucket, then 5 taps fill 5 buckets.

Expectation: C4

Learning Goal: "I can use data to predict an outcome."

Materials: M1_paper_trees, M2_sap_buckets, M5_prediction_chart, M3_modelling_chart

1. Minds On (12 mins): Sap Sam Predicts Buckets

If 1 tree fills 1 bucket of sap, how many buckets for 5 trees? 10 trees?

Bring children to the carpet. Sap Sam in hand. 'Friends — yesterday we collected DATA. Today we use the data to PREDICT.'

2. Action (25 mins): Prediction Stations

- Whole group: predict for 4 more scenarios.
- Send pairs to three stations (~7 min each).
- Stretch challenge at Station C: include problems where predictions might be wrong (variability — some trees give MORE than 1 bucket, some give LESS).
- Discuss the rule.
- Circulate to record formative evidence on the prediction tracker.

3. Consolidation (13 mins): Sam's Predict Brag

Discuss:

- How did you make your prediction?
- What's a rule?
- Can a rule be wrong sometimes?
- What's one prediction you made today?

Exit: Each child writes ONE prediction on a sticky note: 'For ____, I predict ____.'
Notes go on the chart.

Lesson 4: Reflecting and Refining

Goal: We check our model against reality. If it is off, we REFINE it — predictions improve with practice.

Expectation: C4

Learning Goal: "I can check my prediction against what really happens and adjust the model."

Materials: M2_sap_buckets, M5_prediction_chart, M3_modelling_chart, M10_sugar_bush_poster

1. Minds On (12 mins): Check Sap Sam's Predictions

Sap Sam predicted 5 buckets but only got 4. What does that mean?

Bring children to the carpet. Maple + Sap Sam puppets. 'Friends — yesterday Sap Sam PREDICTED. Today we CHECK whether his predictions matched reality.'

2. Action (25 mins): Reflect Stations

- Whole group: predict-and-check 4 more scenarios.
- Send pairs to three stations (~7 min each).
- Stretch challenge at Station C: invent a rule that always works.
- Discuss the cycle.
- Circulate to record formative evidence on the reflection tracker.

3. Consolidation (13 mins): Maple's Refine Brag

Discuss:

- What does REFINE mean?
- Why don't predictions always match reality?
- How does refining make our model better?
- What's a real-life rule you'd refine?

Exit: Each child writes ONE refinement on a sticky note: 'I'd refine my rule to ____.' Notes go on the chart.

Lesson 5: My Real-Life Math

Goal: We can model ANY real-life situation. Pick a topic from your life, ask a question, build a model.

Expectation: C4

Learning Goal: "I can model a real-life situation from my own life."

Materials: M6_capstone_template, M7_situation_cards, M9_syrup_jar_template, M3_modelling_chart

1. Minds On (12 mins): Closing Showcase

Now you model YOUR life. Pick a situation. Walk through all 5 steps.

Bring children to the carpet. Maple + Sap Sam. 'Friends — today you become MATHEMATICAL MODELLERS. Pick a real-life situation from your own life. Walk through the cycle: SITUATION, QUESTION, DATA, PREDICT, REFLECT.'

2. Action (25 mins): Capstone Modelling

- Hand out capstone templates.
- Children work for 12-15 minutes individually.
- Stretch challenge: include a refinement based on imagined or real data.
- Gallery walk: children swap templates and read each other's models.
- Whole-class share: 5-6 children share their model aloud.

3. Consolidation (13 mins): Closing Brag and Certificate Ceremony

Discuss:

- What's the most surprising model you saw today?
- Why do we use the modelling CYCLE (with arrows back)?
- What's something in your life you'll model after this unit?
- What's the difference between data and prediction?

Exit: Each child receives a Junior Mathematical Modeller certificate.

Name: _____

Date: _____

Maple's Sugar Bush Questions

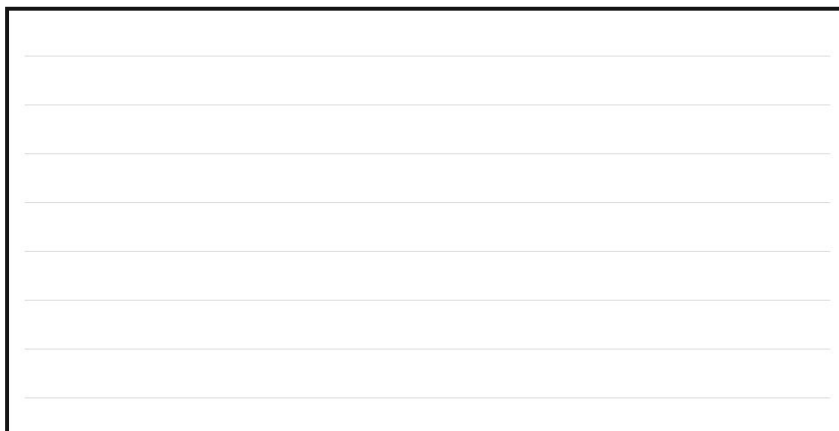
Learning Goal: I can describe a real-life situation and ask math questions about it.

Instructions for the Student:

1. Draw or write what you see in the sugar bush.
2. Write 3 math questions about the sugar bush.
3. Pick something from YOUR day (snack, recess, bus).

The Bush

Draw what you see in the sugar bush:

A large rectangular box with a black border, containing ten horizontal lines for drawing or writing.

Name: _____

Date: _____

Sap Sam's Tree Count

Learning Goal: I can collect data to answer a math question about a real-life situation.

Instructions for the Student:

1. Look at each bush.
2. Use your tree counts from Part 1 to answer these 3 questions.
3. Pick something to count in your day (crayons, books, friends).

Ask the Data

Use the chart to answer 3 questions:

Name: _____

Date: _____

Sap Sam's Predictions

Learning Goal: I can use data to predict an outcome.

Instructions for the Student:

1. Use the rule '1 tree = 1 bucket'.
2. New rule: '1 tree = 2 buckets'.
3. Pick a rule from YOUR day (1 sandwich = 2 slices, 1 friend = 1 hello).

My Rule

Write your own rule and explain:

Name: _____

Date: _____

Maple's Refine Time

Learning Goal: I can check my prediction against what really happens and adjust the model.

Instructions for the Student:

1. Compare 4 predict-vs-actual scenarios.
2. Each rule below was a bit off.
3. Think of a real-life prediction you made.

Refine Your Prediction

Original prediction:

What I noticed:

New prediction:

Name: _____

Date: _____

My Real-Life Model

Learning Goal: I can model a real-life situation from my own life.

Instructions for the Student:

1. Pick a real-life topic.
2. Write a math question about your situation.
3. Collect or imagine data.

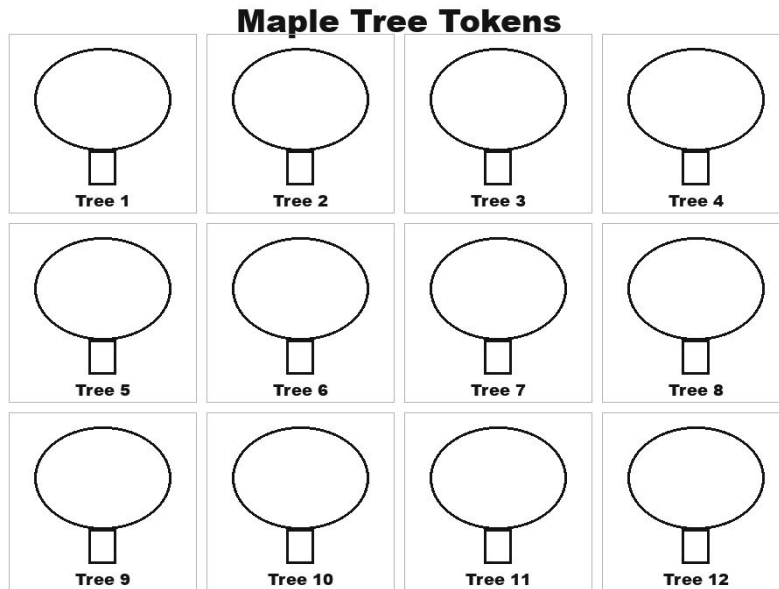
My Topic

Pick a real-life topic:

Paper Maple Trees

(M1_paper_trees)

20 paper tree tokens for the sugar bush counting and modelling.



PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 2

Laminate: yes

USE IN CLASS

Quantity: 20 trees

Lessons: 1, 2, 3

Prep: ~15 min

TEACHER PREP

1. Print 2 pages on cardstock.
2. Laminate.
3. Cut into 20 tokens.
4. Sort into pair sets.
5. Distribute Lessons 1-3.

Paper Sap Buckets

(M2_sap_buckets)

20 paper bucket tokens for collecting sap and matching trees.

Syrup Bucket Tokens



PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 2

Laminate: yes

USE IN CLASS

Quantity: 20 buckets

Lessons: 2, 3, 4

Prep: ~15 min

TEACHER PREP

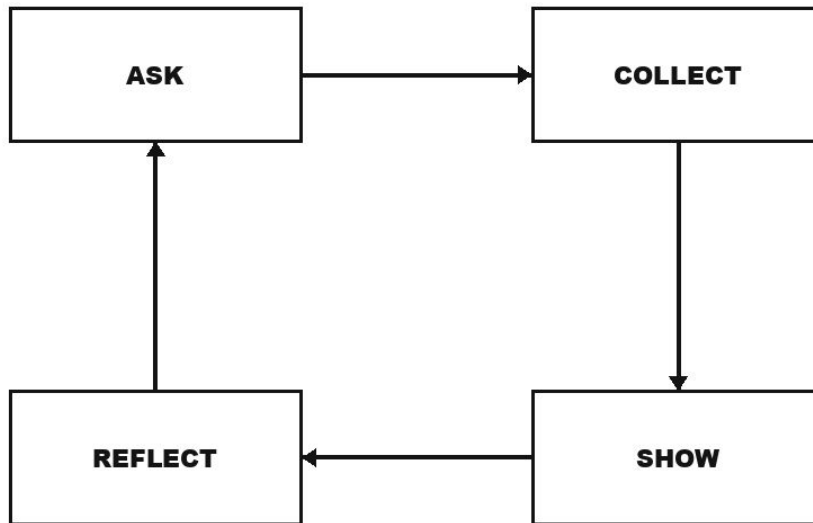
1. Print 2 pages on cardstock.
2. Laminate.
3. Cut into 20 tokens.
4. Sort into pair sets.
5. Distribute Lessons 2-4.

Modelling Cycle Chart

(M3_modelling_chart)

Letter-size chart showing the 5 modelling steps with looping arrows.

Modelling Cycle



PRINT SPECS

Size: letter landscape
Colour: bw
Pages per set: 1
Laminate: yes

TEACHER PREP

1. Print 1 chart per pair.
2. Laminate.
3. Hand out Lesson 1.
4. Use as reference all week.
5. Reuse.

USE IN CLASS

Quantity: 1 per pair plus 1 anchor
Lessons: 1, 2, 3, 4, 5
Prep: ~12 min

Data Collection Table

(M4_data_table)

Letter-size table for recording counts and measurements.

Data Chart

Item	Count	Total

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

USE IN CLASS

Quantity: 1 per pair

Lessons: 2, 3, 4

Prep: ~10 min

TEACHER PREP

1. Print 1 table per pair.

2. Laminate.

3. Hand out Lesson 2.

4. Use dry-erase.

5. Reuse Lessons 3-4.

Prediction Chart

(M5_prediction_chart)

Letter-size chart with PREDICT and ACTUAL columns plus a GAP column.

Predict vs. Actual

Day	Predicted	Actual	Difference

PRINT SPECS

Size: letter landscape
Colour: bw
Pages per set: 1
Laminate: yes

TEACHER PREP

1. Print 1 chart per pair.
2. Laminate.
3. Hand out Lesson 3.
4. Use dry-erase.
5. Reuse Lesson 4.

USE IN CLASS

Quantity: 1 per pair
Lessons: 3, 4
Prep: ~12 min

Capstone Modelling Template

(M6_capstone_template)

Letter-size template for Lesson 5 capstone modelling project.

Capstone Modelling Template	
1	Situation _____
2	Question _____
3	Data _____
4	Predict _____

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: no

USE IN CLASS

Quantity: 1 per child

Lessons: 5

Prep: ~12 min

TEACHER PREP

1. Print 1 per child.

2. Hand out Lesson 5.

3. Children fill in.

4. Collect for assessment.

5. Display gallery.

Real-Life Situation Cards

(M7_situation_cards)

12 cards with everyday situations to model.

Situation Cards

Counted 12 trees on Day 1.	Tapped 8 trees on Day 2.
Filled 5 buckets on Day 3.	Took 3 days to fill the jar.
Friend says the syrup tastes sweetest at the end.	Predict: how many buckets next week?

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 2

Laminate: yes

USE IN CLASS

Quantity: 12 cards

Lessons: 1, 5

Prep: ~15 min

TEACHER PREP

1. Print 2 pages on cardstock.

2. Laminate.

3. Cut into 12 cards.

4. Sort by topic.

5. Distribute Lessons 1 and 5.

Two-Colour Counters

(M8_two_colour_counters)

20 counters per pair for modelling sap collection and tree matching.

Counter Sheet

1	2	3	4	5
6	7	8	9	10
11	12	13	14	15
16	17	18	19	20

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

USE IN CLASS

Quantity: 20 per pair

Lessons: 2, 3, 4

Prep: ~15 min

TEACHER PREP

1. Print 1 sheet per pair on cardstock.

2. Laminate.

3. Cut into 20 counters.

4. Sort into pair bags.

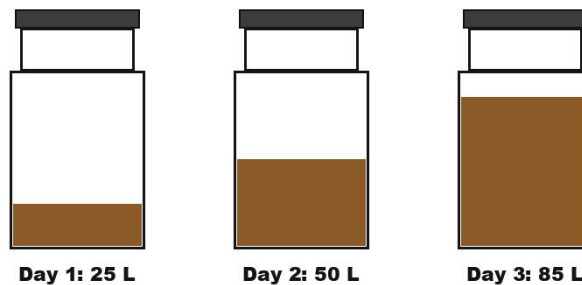
5. Distribute Lessons 2-4.

Syrup Jar Template

(M9_syrup_jar_template)

Printable jar template for the Lesson 5 certificate ceremony.

Syrup Jar — Daily Count



PRINT SPECS

Size: letter portrait
Colour: bw
Pages per set: 1
Laminate: no

TEACHER PREP

1. Print 1 per child.
2. Hand out at certificate ceremony.
3. Children write name and project.
4. Display.
5. Send home.

USE IN CLASS

Quantity: 1 per child
Lessons: 5
Prep: ~8 min

Sugar Bush Anchor Poster

(M10_sugar_bush_poster)

Large classroom anchor showing Maple and Sap Sam at the sugar bush with the modelling cycle.

Maple Sugar Bush

- 1. ASK a question.
- 2. COUNT taps and trees.
- 3. TALLY each day.
- 4. PREDICT the next day.
- 5. REFINE your guess.

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 4

Laminate: yes

USE IN CLASS

Quantity: 1 poster

Lessons: 1, 2, 3, 4, 5

Prep: ~18 min

TEACHER PREP

1. Print 4 pages and tile.

2. Laminate.

3. Tape to wall Day 1.

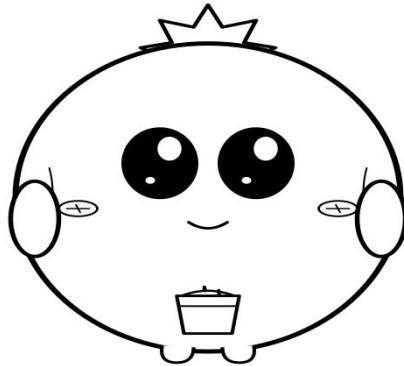
4. Reference all week.

5. Roll for re-use.

Maple Puppet

(char_maple_puppet)

Printable Maple illustration on a popsicle stick for storytelling.



Maple the Maple Tree

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

USE IN CLASS

Quantity: 1 puppet

Lessons: 1, 2, 3, 4, 5

Prep: ~10 min

TEACHER PREP

1. Print on cardstock.

2. Cut.

3. Glue popsicle stick.

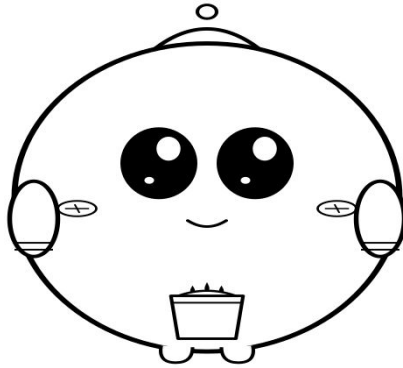
4. Optional: laminate.

5. Store flat.

Sap Sam Puppet

(char_sap_sam_puppet)

Printable Sap Sam illustration on a popsicle stick for storytelling.



Sap Sam the Collector

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

USE IN CLASS

Quantity: 1 puppet

Lessons: 2, 3, 5

Prep: ~10 min

TEACHER PREP

1. Print on cardstock.

2. Cut.

3. Glue popsicle stick.

4. Optional: laminate.

5. Store flat.

Name: _____

Date: _____

Summative Assessment Rubric

Expectation	Level 1 (Beginning)	Level 2 (Developing)	Level 3 (Achieving)	Level 4 (Exceeding)
C4 apply the process of mathematical modelling to represent, analyse, make predictions, and provide insight into real-life situations	Cannot identify a real-life situation or ask a math question.	With prompts, walks through 2-3 of the 5 modelling steps.	Independently walks through all 5 modelling steps with a coherent topic.	Walks through all 5 steps with a coherent topic and includes a refinement based on the data.

Junior Mathematical Modeller Certificate

This certificate is proudly presented to:

For successfully completing the 5-Day Real-Life Modelling unit and demonstrating the skills of a Junior Mathematical Modeller. Maple and Sap Sam congratulate the student.

By completing this unit, this student has demonstrated:

- Describing a real-life situation
- Asking a math question
- Collecting data
- Making a prediction
- Reflecting and refining the model

Marketplace Listing

A 5-day Ontario Grade-1 unit on mathematical modelling (C4) framed around the Canadian sugar bush — Maple the Maple Tree and Sap Sam the Collector.

Price: \$9.99 CAD

Pages: ~32

Classroom time: 250 min

Teacher prep: ~90 min

Top tags: grade-1, math, ontario-curriculum, modelling, algebra, real-life, canadian, sugar-bush

What's included:

- 5 fully-scripted lesson plans with 50-minute durations and 250 minutes of total teaching time
 - 5 student worksheets aligned to each lesson
 - 1 mid-unit formative assessment (Modelling Check-In)
 - 10 printable teacher manipulatives plus 2 character puppets: paper maple trees (20), sap buckets (20), modelling cycle chart, data collection table,...
- 1 end-of-unit reflection sheet with 5 self-assessment prompts
- 4 class observation trackers: 1 diagnostic + 3 formative
- 1 single-row summative rubric covering C4 across 4 levels
- 1 summative-administration guide tying the rubric to Lesson 5 capstone
- Junior Mathematical Modeller certificate (one per child)
- Detailed expanded walkthroughs covering common misconceptions
- Differentiation tracks built into every lesson